Soluciones Certamen 3

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Problem 1 - Exam 3

1. (**7 points**) The transformation of coordinates and momenta has a form

$$P = \frac{\alpha p^2 q}{\sqrt{1 + p^2 q^2}}, \qquad Q = \beta \frac{\sqrt{1 + p^2 q^2}}{p}, \qquad \alpha, \beta = \mathrm{const}$$

   (**3 points**) Find the values of constants α, β for which the transformation is canonical. For these values of α, β, find explicitly the generating function F which corresponds to the transformation.

   (**4 points**) Apply the transformation found in the previous item to the hamiltonian of harmonic oscillator $H(p, q) = \frac{p^2}{2m} + \frac{kq^2}{2}$ and find the hamiltonian $\tilde{H}(P, Q)$ in terms of the new variables (P, Q). Write out the canonical equations in terms of these new variables.

Problem 1 - part 1

Fastest check that transformation is canonical:

Check for what values α, β the identities

$$[P,\, Q]_{p,q} = 1, \quad [Q,\, Q]_{p,\,q} = 0, \quad [P,\, P]_{p,q} = 0,$$

are satisfied identically (clearly, the last 2 identities are always satisfied in 1D, need to check only the first)
−Evaluation of Poisson brackets using conventional rules:

$$[P,\, Q] = \frac{\partial P}{\partial p}\frac{\partial Q}{\partial q} - \frac{\partial P}{\partial q}\frac{\partial Q}{\partial p} = \alpha\beta$$

Conclude that the transformation is canonical when $\alpha\beta = 1$, $\beta = 1/\alpha$. We cannot fix completely value of α, it plays the role of rescaling parameter

Problem 1 - part 1

Some observations:

Assume $\alpha = \beta = 1$ in this slide for simplicity:

$$P = \frac{p^2 q}{\sqrt{1 + p^2 q^2}}, \qquad Q = \frac{\sqrt{1 + p^2 q^2}}{p}, \tag{1}$$

$$PQ = pq, \qquad \frac{P}{p} = \frac{q}{Q}$$

$$q = \frac{PQ^2}{\sqrt{P^2 Q^2 + 1}}, \quad p = \frac{\sqrt{P^2 Q^2 + 1}}{Q} \tag{2}$$

To get inverse transformation (2), just replace $p \leftrightarrow Q$, $q \leftrightarrow P$ in direct transformation (1).

Problem 1 - part 1

Now we'll find the generating function

►There are 4 generating functions $F_{1\ldots 4}$, all are related

−Let's for definiteness work with $F_1(q, Q)$

− We'll rewrite canonical transformation as

$$p = \frac{1}{\sqrt{\alpha^2 Q^2 - q^2}} = \frac{\partial F}{\partial q}, \quad P = -\frac{\partial F}{\partial Q} = \frac{q}{Q\sqrt{\alpha^2 Q^2 - q^2}}$$

Can check that system is compatible,

$$\frac{\partial^2 F}{\partial q\,\partial Q} = \frac{\partial^2 F}{\partial Q\,\partial q} = -\frac{\alpha^2 Q}{(\alpha^2 Q^2 - q^2)^{3/2}}$$

►Solve equations integrating it directly one-by-one

$$F = \int dq\,\frac{\partial F}{\partial q} = \arctan\left(\frac{q}{\sqrt{\alpha^2 Q^2 - q^2}}\right) + a\,(Q)$$

$$-\frac{\partial F}{\partial Q} = \frac{q}{Q\sqrt{\alpha^2 Q^2 - q^2}} + \frac{\partial a(Q)}{\partial Q} = \left(P = \frac{q}{Q\sqrt{\alpha^2 Q^2 - q^2}}\right) \Rightarrow a = \mathrm{const} = 0.$$

$$\Rightarrow F = \arctan\left(\frac{q}{\sqrt{\alpha^2 Q^2 - q^2}}\right)$$

Since canonical transformation and its generating function do not depend on t explicitly, $\tilde{H} = H$, i.e. should merely substitute argument in hamiltonian. For this need inverse transform $q = q(Q, P)$ and $p = p(Q,\ P)$:

$$q = \frac{\alpha P Q^2}{\sqrt{P^2 Q^2 + 1}}, \quad p = \frac{\sqrt{P^2 Q^2 + 1}}{\alpha Q}$$

►Hamiltonian:

$$\tilde{H}\left(P,\ Q\right) = H\left(p\left(P,\ Q\right),\ q\left(P,\ Q\right)\right) = \frac{\alpha^2 k P^2 Q^4}{2P^2 Q^2 + 2} + \frac{P^2 Q^2 + 1}{2\alpha^2 m Q^2}$$

►Canonical equations:

$$\dot{P} = -\frac{\partial \tilde{H}\left(P,\ Q\right)}{\partial Q} = \frac{1}{m Q^3 \alpha^2} - \frac{\alpha^2 k P^2 Q^3 \left(P^2 Q^2 + 2\right)}{\left(P^2 Q^2 + 1\right)^2},$$

$$\dot{Q} = \frac{\partial \tilde{H}\left(P,\ Q\right)}{\partial P} = \frac{P}{m\alpha^2} + \frac{\alpha^2 k P Q^4}{\left(1 + P^2 Q^2\right)^2}$$

Problem 2

2. (**8** ***points***) A nonrelativistic charged particle of mass m is constrained to move in a horizontal plane, in the field of the harmonic oscillator $U = kr^2/2$ and the constant magnetic field $\boldsymbol{B}$ perpendicular to the plane. The vector potential $\boldsymbol{A}$ may be chosen as

$$\boldsymbol{A} = \frac{1}{2}\boldsymbol{B} \times \boldsymbol{r}.$$

Analyze this system in polar coordinates:

(3 points) Write out the lagrangian, the generalized momenta and construct the hamiltonian

(2 points) Write out the canonical equations of motion.

(3 points) Write out the equation of Hamilton-Jacobi and integrate it using separation of variables.

Note: *The goal of this part is to test your knowledge of the Hamilton-Jacobi formalism, for this reason solution with any other method will not be taken into account.*

Lagrangian (see Lecture 12; in polar coordinates $\boldsymbol{v} = \dot{r}\hat{\boldsymbol{r}} + r\dot{\varphi}\hat{\boldsymbol{\varphi}} + \dot{z}\hat{\boldsymbol{z}}$):

$$L = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) - \frac{kr^2}{2} + q\,\underbrace{\frac{Br^2}{2}\dot{\varphi}}_{\boldsymbol{v}\cdot\boldsymbol{A}}, \qquad \boldsymbol{A} = \frac{1}{2}\boldsymbol{B}\times\boldsymbol{r} = \frac{Br}{2}\hat{\boldsymbol{\varphi}}$$

►Generalized momenta:

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}, \quad p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = m\,r^2\dot{\varphi} + q\frac{Br^2}{2},$$

−no dependence on φ, cyclic variable, $p_\varphi = \mathrm{const}$

► Construct hamiltonian using Legendre transform:

$$H \equiv \sum_a p_a\dot{q}_a - L = p_r\dot{r} + p_\varphi\dot{\varphi} - L = \frac{p_r^2}{2m} + \frac{\left(p_\varphi - q\frac{Br^2}{2}\right)^2}{2mr^2} + \frac{kr^2}{2}$$

►Canonical equations:

$$\dot{p}_r = -\frac{\partial H}{\partial r} = -\frac{\left(q\frac{B}{2}\right)^2 r^4 - p_\varphi^2}{mr^3} - kr, \quad \dot{p}_\varphi = -\frac{\partial H}{\partial \varphi} = 0,$$

$$\dot{r} = \frac{\partial H}{\partial p_r} = \frac{p_1}{m}, \quad \dot{\varphi} = \frac{\partial H}{\partial p_\varphi} = \frac{\left(p_\varphi - q\frac{Br^2}{2}\right)}{m}.$$

Problem 2 - Hamilton-Jacobi formalism

$$\frac{\partial S}{\partial t} + H\left(p_a = \frac{\partial S}{\partial q_a},\, q_a\right) = 0$$

►Energy is conserved since H does not depend on time
►Apply separation of variables: effectively assume that S has a form

$$S = -Et + a(r) + b(\varphi)$$

$$E = \frac{(a'(r))^2}{2m} + \frac{\left(b'(\varphi) - q\frac{Br^2}{2}\right)^2}{2mr^2} + \frac{kr^2}{2}$$

$$2mE = \left(a'(r)\right)^2 + \frac{1}{r^2}\left(b'(\varphi) - q\frac{Br^2}{2}\right)^2 + kmr^2$$

We can see that φ-dependence appears only in $b'(\varphi)$, so must conclude that

$$b'(\varphi) = p_\varphi = \mathrm{const}$$

Problem 2

$$\left(\frac{da}{dr}\right)^2 = 2mE - kmr^2 - \frac{1}{r^2}\left(p_\varphi - q\frac{Br^2}{2}\right)^2$$

$$\Rightarrow a(r) = \int^r dr \sqrt{2mE - kmr^2 - \frac{1}{r^2}\left(p_\varphi - q\frac{Br^2}{2}\right)^2}$$

$$= \int^r dr \sqrt{2m\left(E + \frac{qB}{2m}p_\varphi\right) - m\left(k + \frac{q^2B^2}{4m}\right)r^2 - \frac{p_\varphi^2}{r^2}}$$

►Same integral as for plane harmonic oscillator in polar coordinates
– but with "adjusted" energy $\tilde{E} = E + \frac{qB}{2m}p_\varphi$ and elasticity $\tilde{k} = k + \frac{q^2B^2}{4m}$
– i.e. the magnetic field just changes parameters
(Mathematica returns the result in terms of elementary functions but too lengthy)

$$\text{action}: \qquad S = -Et + a(r) + p_\varphi \varphi$$

►Now, after we got S, we can extract trajectories from

$$\frac{\partial S}{\partial E} = \text{const} = -t_0, \qquad \frac{\partial S}{\partial p_\varphi} = \text{const} = -\varphi_0.$$